

Biological Degradation and Greenhouse Gas Emissions during Pre-Storage of Liquid Animal Manure

Henrik B. Møller,* Sven G. Sommer, and Birgitte K. Ahring

ABSTRACT

Storage of manure makes a significant contribution to global methane (CH_4) emissions. Anaerobic digestion of pig and cattle manure in biogas reactors before outside storage might reduce the potential for CH_4 emissions. However, manure pre-stored at 15 to 20°C in buildings before anaerobic digestion may be a significant source of CH_4 and could reduce the potential CH_4 production in the biogas reactor. Degradation of energy-rich organic components in slurry and emissions of CH_4 and carbon dioxide (CO_2) from aerobic and anaerobic degradation processes during pre-storage were examined in the laboratory. Newly mixed slurry was added to vessels and stored at 15 and 20°C for 100 to 220 d. During storage, CH_4 and CO_2 emissions were measured with a dynamic chamber technique. The ratio of decomposition in the subsurface to that at the surface indicated that the aerobic surface processes contributed significantly to CO_2 emission. The measured CH_4 emission was used to calculate the methane conversion factor (MCF) in relation to storage time and temperature, and the total carbon-C emission was used to calculate the decrease in potential CH_4 production by anaerobic digestion following pre-storage. The results show substantial methane and carbon dioxide production from animal manure in an open fed-batch system kept at 15 to 20°C, even for short storage times, but the influence of temperature was not significant at storage times of <30 d. During long-term storage (90 d), a strong influence of temperature on the MCF value, especially for pig manure, was observed.

ANTHROPOGENIC EMISSIONS of the greenhouse gases CO_2 and CH_4 have increased significantly during the 20th century (Husted, 1994; Steed and Hashimoto, 1994; Sommer et al., 2002). Relative to CO_2 , the amounts of CH_4 in the atmosphere are low, but the global warming potential (GWP) of CH_4 is 21 times higher than that of CO_2 on a molecular basis (Steed and Hashimoto, 1994). Recent concerns about effects of the gaseous release of hydrocarbons on the degradation of the ozone layer and global warming have led to increased awareness of the release of CH_4 from animal waste production facilities (Hill et al., 2001).

During storage of manure, CH_4 and CO_2 are formed by anaerobic bacterial degradation of organic matter (Steed and Hashimoto, 1994). Carbon dioxide is also formed by aerobic bacterial degradation (Patni and Jui, 1985). However, to our knowledge, few (if any) studies have quantified the proportion of aerobic and anaerobic processes in liquid manure storage systems. The distinc-

tion between anaerobic and aerobic processes is important, since CO_2 released by the degradation of organic matter in manure does not contribute to global warming, because the CO_2 produced has previously been absorbed from the air and metabolized by plants, which were used for feeding the animal, thus being part of carbon cycling and not a source of additional CO_2 (Külling et al., 2002). On the other hand, CH_4 released by anaerobic degradation contributes significantly to global warming.

Protocols for estimating CH_4 emission from manure have been set out by the Intergovernmental Panel on Climate Change (1996). The protocols estimate the ultimate CH_4 production from anaerobic digestion of organic material as B_0 (Intergovernmental Panel on Climate Change, 1996), and the CH_4 emission in a given manure management system and climate is calculated by multiplying B_0 by a methane conversion factor (MCF). The MCF is affected by several factors such as storage time, manure characteristics, amount of methanogenic inoculum in the storage after emptying, time and temperature distribution between indoor and outdoor storage, daily temperature variations, and seasonal temperature variations (Intergovernmental Panel on Climate Change, 1996).

In Western Europe, manure is stored in buildings for a period that varies between 15 and 60 d at temperatures of 15 to 20°C (Sommer et al., 2002). Slurry from the animal houses is flushed to an outside concrete store and is stored in the open from a few months up to 9 to 10 mo, depending on the duration of the winter (Burton, 1997). Intergovernmental Panel on Climate Change (1996) protocols estimate the MCF value for manure stored in animal houses in cool regions at 5% if the manure is stored for less than 30 d, and 10% if the manure is stored for more than 30 d. Several studies have estimated CH_4 emissions during storage of manure at temperatures between 10 and 30°C (Husted, 1994; Zeeman, 1994; Steed and Hashimoto, 1994). In all studies, the degradation processes are assumed to be mainly anaerobic. Steed and Hashimoto (1994) examined emissions in both open and closed vessels and found that the MCF in open vessels was only 18% of the MCF in closed vessels when stored at 20°C. From this, they assumed that the oxygen in the air was inhibiting methanogenesis.

This study examines the degradation of organic components and CH_4 production in slurry pits inside animal houses at 15 to 20°C and the production of CO_2 and CH_4 at the surface of liquid and in the subsurface of the slurry. In the study, feces, urine, and water were

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Abbreviations: MCF, methane conversion factor; VFA, volatile fatty acids; VS, volatile solids.

Table 1. Composition of the raw manure, the inoculum, and the mixture of raw manure (84.8%), inoculum (7.6%), and water (7.6%) used in the experiment (20°C).

Fraction†	Type of manure					
	Pig			Cattle		
	Feces and urine	Inoculum	Complete slurry	Feces and urine	Inoculum	Complete slurry
B_0 , L CH ₄ g ⁻¹ VS	0.35			0.25		
B_{100} , L CH ₄ g ⁻¹ VS	0.508	0.469	0.507	0.485	0.485	0.485
B_0/B_{100}	0.70			0.50		
DM, g L ⁻¹	95.5	65.5	86.0	103.2	100.8	95.2
COD, g L ⁻¹	50.7	28.8	49.4	49.4	19.9	56.5
VS, g kg ⁻¹ DM	836.5	731.3	830.4	793.1	740.1	788.8
NH ₄ ⁺ -N, g kg ⁻¹ DM	25.7	41.9	26.6	19.0	18.5	18.9
Total N, g kg ⁻¹ DM	70.1	64.1	69.7	49.4	40.7	48.7
Protein‡, g kg ⁻¹ DM	194.5‡	138.8	191.3	212.3§	138.8	205.4
Lipid, g kg ⁻¹ DM	100.8	45.3	97.5	60.9	65.8	61.3
Carbohydrates¶, g kg ⁻¹ DM	541.2	547.3	541.5	520.9	535.6	522.1
Average CH ₄ emission rate after 30 d, mg kg ⁻¹ slurry d ⁻¹ ± SD						
15°C			16.4 ± 1.06			11.6 ± 0.66
20°C			24.7 ± 4.49			14.7 ± 1.47
Average CH ₄ emission rate after 100 d, mg kg ⁻¹ slurry d ⁻¹ ± SD						
15°C			9.3 ± 0.91			5.5 ± 0.41
20°C			55.1 ± 9.6			6.4 ± 1.39

† B_0 , methane emission; B_{100} , methane yield; COD, chemical oxygen demand; DM, dry matter; VS, volatile solids.

‡ Protein (pig feces + urine) = $6.25 \times \text{total N} \times 0.29 + 2.14 \times 0.71 \times (\text{total N}/\text{NH}_4^+\text{-N})$. It is assumed that 71% of N is excreted as N in urine (urea and NH₄⁺-N) (Poulsen et al., 2001).

§ Protein (cattle feces + urine) = $6.25 \times \text{total N} \times 0.6 + 2.14 \times 0.4 \times (\text{total N}/\text{NH}_4^+\text{-N})$. It is assumed that 40% of N is excreted as N in urine (urea and NH₄⁺-N) (Poulsen et al., 2001).

¶ Carbohydrates = VS - protein - lipid.

added to an open incubation vessel with inoculum present for 1 to 10 d; thereafter the slurry was incubated until 100 and 200 d after initiation of the experiment. Emission of CH₄ and CO₂ from the slurry was measured with a dynamic chamber technique. The experiment thus simulated degradation of organic components and gas emission during storage in animal houses and for a longer period in stores outside, simulating a summer period.

MATERIALS AND METHODS

Experimental Setup of Fed-Batch Digester

In each experiment, three vessels (height = 1 m, diameter = 0.10 m) containing slurry were incubated at $20 \pm 0.5^\circ\text{C}$ and three vessels at $15 \pm 0.5^\circ\text{C}$. Cattle and pig slurry (Table 1) were incubated at both temperature regimes. The vessels were gradually filled to 22 cm from the top of the vessel by adding a mixture of feces and urine (84.8%), water (7.6%), and inoculum (7.6%) over a period of 10 d, corresponding to the normal time taken to fill a manure channel under a slatted floor. All of the inoculum was added at the beginning of the experiment. The feces and urine had been collected immediately after excretion and stored at -18°C until the time of addition to the vessel. The pigs producing the urine and feces were fed a diet containing (dry matter basis) 24% soybean meal, 49.8% barley, 20% wheat, 1% molasses, 2% fat, and 3.2% minerals. The dairy cattle producing the urine and feces were fed a diet containing (dry matter basis) 21.6% concentrates, 55.2% roughage, 22.8% barley, and 0.4% minerals. The inoculum consisted of pig or cattle manure stored for 30 d at 15 or 20°C , respectively, allowing a population of anaerobic bacteria to adapt to the environment. The vessels were incubated in a thermo-controlled water bath at constant temperatures of 15 or 20°C , respectively.

At the beginning and end of the experiment, representative

samples of the manure were taken after vigorous stirring. The samples were analyzed for the following important parameters: dry matter, volatile solids (VS), total nitrogen, NH₄-N, volatile fatty acids (VFA), pH, and lipids.

Gas Emission

From the start of filling the vessels, gas emissions were measured continuously with dynamic chambers, giving the emissions from both the subsurface and surface of the slurry. After the vessels had been filled, 17.9% of the slurry surface was covered with an additional static chamber using a tube submerged 2 cm into the slurry and measuring gas emissions primarily from the subsurface of the slurry (Fig. 1).

The dynamic chambers were set up by closing the vessels containing slurry with gas-impermeable lids, leaving an open headspace between the slurry surface and the lids for measuring gas emissions. Input and output ports on opposite sides of the top of the vessel directed air into the headspace. The air was sucked into each headspace by a pump at an airflow rate of 1.1 L min^{-1} . The concentrations of CH₄ and CO₂ in the incoming and exhaust air were analyzed with an on-line multigas monitor (photoacoustic monitor; Bruel & Kjaer, Naerum, Denmark). The quality of the measurements was checked by taking air samples and measuring CH₄ and CO₂ with gas chromatography. From the gas flow and the gas concentrations, mass balances for the losses of CH₄ and CO₂ were calculated.

After the columns had been filled with material an additional static chamber was inserted, consisting of cylindrical tubes (length = 0.75 m, diameter = 4.4 cm) covering 17.9% of the surface and submerged 2 cm from the liquid surface and closed with a lid. The tubes were inserted through a cylindrical hole in the lid of the dynamic chamber, fitting closely to the lid. A stopper was inserted into the top of the tube. The static chamber was evacuated with a syringe until the headspace were completely filled with slurry. Gas emission was calculated as the gas concentration $\times$ the volume of gas

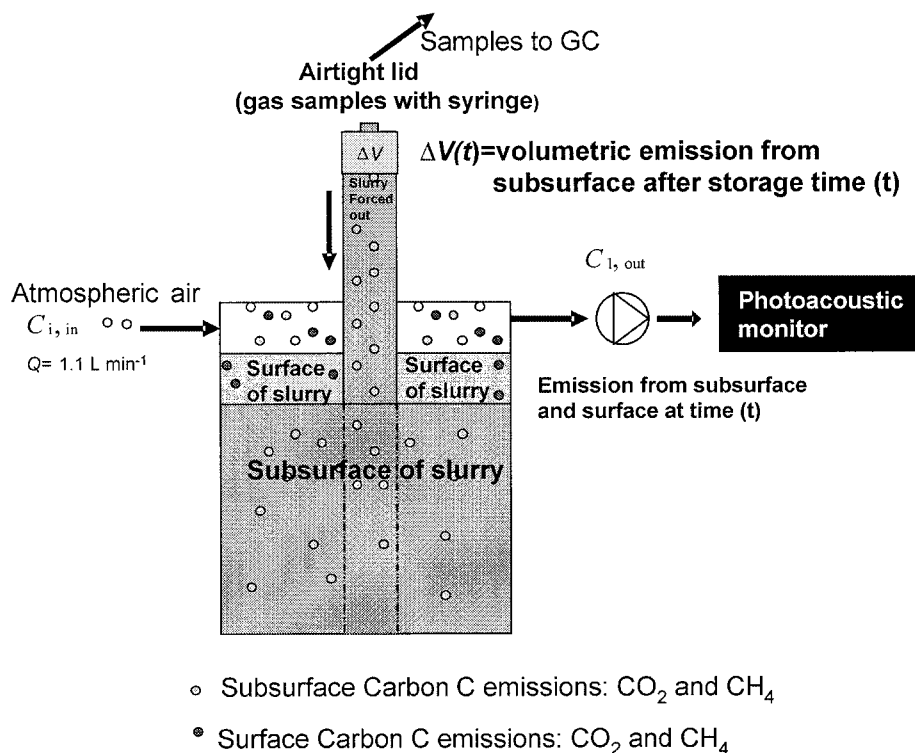


Fig. 1. Schematic diagram of the experimental setup. The dimensions of the outer tube (dynamic chamber) were 104 mm × 100 cm and the dimensions of the inner tube (passive chamber) were 44 mm × 75 cm. The term Q is the airflow through the vessel.

produced. The volume of gas produced at a given time interval was determined as the amount of gas taken out with a syringe to refill the static chamber with slurry. The static chambers were evacuated for gas when the gas had displaced more than half of the slurry. The gas samples were analyzed for CO_2 and CH_4 with gas chromatography.

Methane was measured on a Hewlett-Packard (Palo Alto, CA) 5890 Series II gas chromatograph with a flame ionization detector. Methane was isolated with a 1.83-m × 3.1-mm column with Poropak N 80/100. Helium was used as carrier gas at 30 mL min⁻¹ and the temperatures of injection port, oven, and detector were 110, 40, and 270°C, respectively. Carbon dioxide was measured on a Varian (Palo Alto, CA) 3350 gas chromatograph equipped with a thermal conductivity detector. It was equipped with a 1-m × 3.1-mm column with a 2-m × 3.1-mm Haysep R 80/100 molecular sieve (Valco Instruments, Houston, TX). The carrier gas was He at a flow rate of 30 mL min⁻¹ and the temperatures of oven and detector were 40 and 150°C, respectively.

Manure Analysis

The total N content was analyzed by the Kjeldahl method and a Kjellfoss (Copenhagen, Denmark) 16200. The dry matter content was determined after a 24-h drying period at 105°C. The ammonium (NH_4^+) in the manure was analyzed by a QuickChem 4200 flow injection analyzer (Lachat Instruments, Milwaukee, WI). The manure pH was determined with a pH meter (Radiometer A/S, Copenhagen, Denmark). The chemical oxygen demand (COD) was analyzed colorimetrically (Spektroquant Nova 60; Merck, Darmstadt, Germany) in accordance with a method described by the American Public Health Association (1995). The concentration of VFA $\text{C}_2\text{--C}_5$ was determined by means of a gas chromatograph (Hewlett-Packard 6850A). The column was an HP INNOWax, 30 m × 0.25 mm × 0.25 μm. The carrier gas was He. The temperature

of the column was gradually increased from 110 to 220°C at a rate of 10°C min⁻¹.

Methane Production Analysis

Values for B_0 were determined in a batch test. The methods used are described in international standard ISO 11734 (International Organization for Standardization, 1995). The test medium was feces taken directly after excretion from the pigs and cattle and frozen until used in the batch. Inoculum from a farm-scale biogas plant was used, which had been kept for two weeks before the test at 35°C to remove most of the remaining methane production. Each test medium was digested in triplicate and the gas production was subtracted from the gas production in the control medium (inoculum and water only).

Calculations

Carbon dioxide and CH_4 emission rates (E_i , L kg⁻¹ VS_{initial} d⁻¹), determined with the dynamic chamber [$E_1(t)$] and the static chamber [$E_2(t)$], respectively, were described by the following equations:

$$E_1(t) = \frac{(\bar{C}_{1,\text{out}} - \bar{C}_{1,\text{in}}) \times Q}{M_{\text{initial}} \times K_1(t) \times (100 - K_2)/100} \quad [1]$$

$$E_2(t) = \frac{[C_{2,\text{headspace}}(t)] \times \Delta V_{\text{headspace}}(t)}{\Delta t \times M_{\text{initial}} \times K_1(t) \times (K_2/100)} \quad [2]$$

where $\bar{C}_{1,\text{out}}$ is the concentration of CH_4 or CO_2 (l/l) in the air leaving the vessels, $\bar{C}_{1,\text{in}}$ is the concentration of CH_4 or CO_2 (l/l) in the background air, and Q is the airflow through the vessel (L d⁻¹). The term M_{initial} is the total amount of volatile solids added to the vessels, $K_1(t)$ is a correction factor for the accumulated amount of material removed from the columns at time t , and K_2 is a constant for the surface area covered

with a tube (17.89%) for collection of gas under the surface by use of the static chamber method. The term $C_{2, \text{headspace}}$ is the concentration of CH_4 or CO_2 (l/l) in the headspace of the tube, and $\Delta V_{\text{headspace}}(t)$ is the volume of gas produced by use of the static chamber method in period Δt .

The accumulated emission of carbon (total E) at time (t) as a fraction of carbon initially present ($\text{g C-CO}_2 \text{ g}^{-1} C_{\text{slurry}}$ and $\text{g C-CH}_4 \text{ g}^{-1} C_{\text{slurry}}$) was calculated from summation of Eq. [3] and [4]:

$$\text{total } E_{\text{CO}_2} = \sum_{t=t}^{t=t+1} [E_{1, \text{CO}_2}(t) + E_{1, \text{CO}_2}(t+1)] \times \frac{\Delta t \times \text{MW}_C}{(V_{20} \times M_C)} \quad [3]$$

$$\text{total } E_{\text{CH}_4} = \sum_{t=t}^{t=t+1} [E_{1, \text{CH}_4}(t) + E_{1, \text{CH}_4}(t+1)] \times \frac{\Delta t \times \text{MW}_C}{(V_{20} \times M_C)} \quad [4]$$

where $E_{1, \text{CO}_2}(t)$ and $E_{1, \text{CH}_4}(t)$ are the emission of carbon dioxide and methane at time t (L d^{-1}); MW_C is the molar weight of carbon (12 g mol^{-1}); and M_C is the amount of C (kg) per kg VS added to the vessels, with an assumption of an average composition of lipid ($\text{C}_{57}\text{H}_{104}\text{O}_6$), protein ($\text{C}_5\text{H}_7\text{O}_2\text{N}$), and carbohydrate ($\text{C}_6\text{H}_{10}\text{O}_5$) (Angelidaki et al., 1999). The term V_{20} is the volume of 1 mol of gas at 20°C and 0.101 MPa pressure (20.87 L).

The percentage loss in the theoretical methane yield (B_u) calculated from Bushwell's formula (Symons and Bushwell, 1933) during storage of manure is assumed to be closely correlated with the ratio of carbon initially present that is released as CH_4 and CO_2 emission. However, since only a fraction of the carbon present is degradable, both the percentage loss of CH_4 that can be retrieved by anaerobic digestion of the manure subsequent to the pre-storage and the MCF value need to be multiplied by the inverse of the ultimate CH_4 yield (B_o). The degradability of carbon can be calculated as the ratio of the ultimate CH_4 yield (B_o) and the theoretical CH_4 (B_u) calculated from Bushwell's formula. Thus, the percentage loss of the potential CH_4 yield of the manure subsequent to storage [$L_{B_o(t)}$] was calculated by means of Eq. [5] and the MCF value was calculated by Eq. [6]:

$$L_{B_o}(t) = \frac{(\text{total} - E_{\text{CO}_2} + \text{total} - E_{\text{CH}_4}) \times \frac{B_u}{B_o} \times 100}{\text{total} - E_{\text{CH}_4}} \quad [5]$$

$$\text{MCF} = \left(\frac{\text{total} - E_{\text{CH}_4} \times V_{20}}{\text{MW}_C \times B_o} \right) \times 100 \quad [6]$$

RESULTS AND DISCUSSION

Effect of Storage Time and Temperature on the pH and Volatile Fatty Acid Content

At low temperatures methane is naturally produced by anaerobic digestion of organic matter (Steed and Hashimoto, 1994; Zeeman, 1994; Nozhevnikova et al., 2000). The anaerobic conversion of organic matter leads to the formation of intermediate VFAs—the main ones being acetate, propionate, and butyrate—where acetate is produced by syntrophic metabolism of butyrate and propionate. Acetate is then further degraded to CO_2 and CH_4 by methanogens. Hydrogen produced during this process is consumed by hydrogenotrophic methanogens, contributing to the production of CH_4 . Accumula-

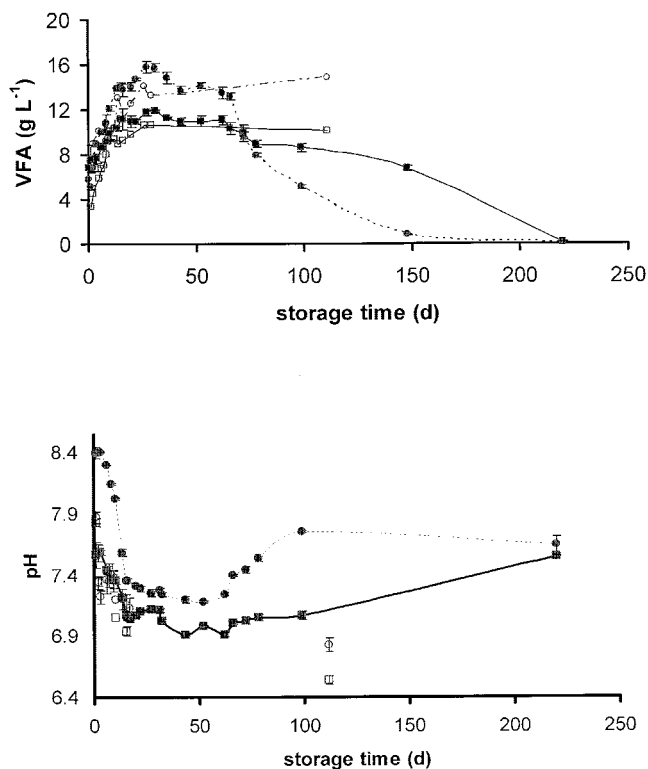


Fig. 2. Volatile fatty acids (VFA) content and pH in columns with mixtures of feces, urine, and inoculum kept at 15 and 20°C , respectively. Open squares, cattle manure stored at 15°C ; open circles, pig manure stored at 15°C ; solid squares, cattle manure stored at 20°C ; solid circles, pig manure stored at 20°C . Error bars represent the standard deviation.

tion of acetate, and particularly hydrogen, leads to the inhibition of VFA degradation (Nozhevnikova et al., 2000).

Figure 2 shows that the total content of VFA in manure increases within the first week after adding slurry to the vessels. The level of total VFA is higher in pig manure than in cattle manure, with a maximum at about 16 g L^{-1} when kept at 20°C and a maximum of 18 g L^{-1} when kept at 15°C . During the initial 90 d, acetic acid comprised more than 60% of the VFA in both cattle and pig manure incubated at 20°C , whereas in columns kept at 15°C , less than 60% of the VFA consisted of acetic acid (Fig. 3). At 15°C the acetate to propionate ratio was lower than at 20°C during the first 100 d, which corresponds with findings of Nozhevnikova et al. (2000). The relative content of propionate increased during the first 100 d of storage at both 15 and 20°C . The increase in propionate in stored pig manure was higher than in cattle manure. The increase in propionate indicates that, at low temperatures, propionate accumulates as an intermediate product because it is degraded at a lower rate than butyrate and acetate (Nozhevnikova et al., 2000).

In all columns, the pH increased during the first few days after the slurry application due to degradation of urea to ammonia then the pH declined because acetogenic bacteria produced acid pH buffers in the form of VFA and CO_2 (Paul and Beauchamp, 1989). The pH increased again after 50 d when the manure was kept

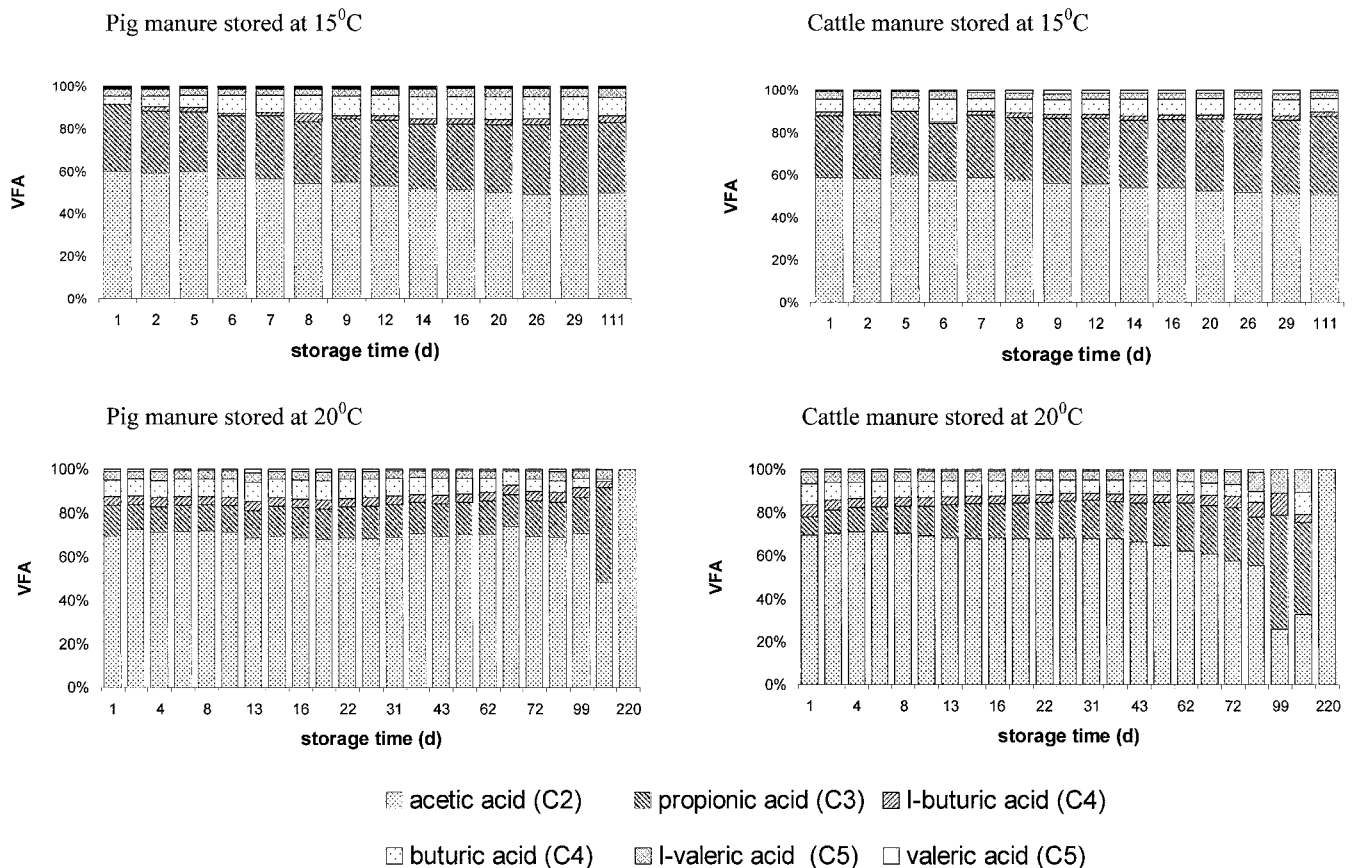


Fig. 3. Distribution of the volatile fatty acids (VFA) content in pig and cattle manure at different storage times kept at 15 and 20°C, respectively.

at 20°C, while pH decreased even further after more than 100 d when the manure was kept at 15°C. The different pattern of pH values at high and low temperatures reflects the fact that the concentrations in acid buffers remain high throughout the experiment in slurry at 15°C (Fig. 2).

Emission of Gases

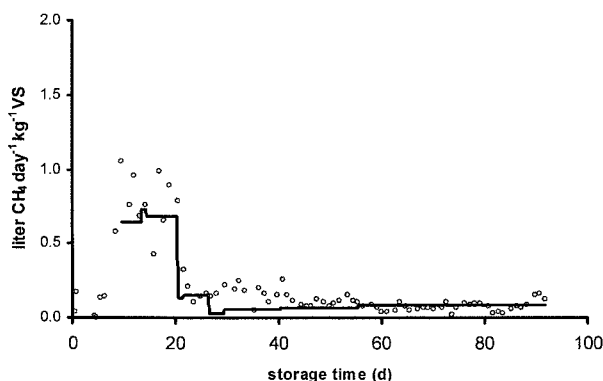
Losses of carbon from animal manure during storage in fed-batch systems are a consequence of degradation by aerobic or methanogenic bacteria, while losses due to VFA volatilization are unlikely because of the insignificant amounts of un-ionized VFA that exist at pH values of >7, which is the most common pH in slurries (Sommer and Husted, 1995). Degradation of organic components to CO₂ by aerobic bacteria and to CH₄ and CO₂ through anaerobic degradation will lead to emissions of CO₂ and CH₄.

In this study, the carbon emitted by both aerobic and anaerobic processes was estimated by the dynamic chamber technique, measuring CH₄ and CO₂ emissions from both the surface and the subsurface phases. Emissions of gases produced by anaerobic processes were estimated by the static chamber technique (subsurface phase), measuring CH₄ and CO₂ emissions 20 mm below the surface. If the emissions calculated with the dynamic chamber and static chamber are within the same range, the degradation will be predominantly anaerobic. On

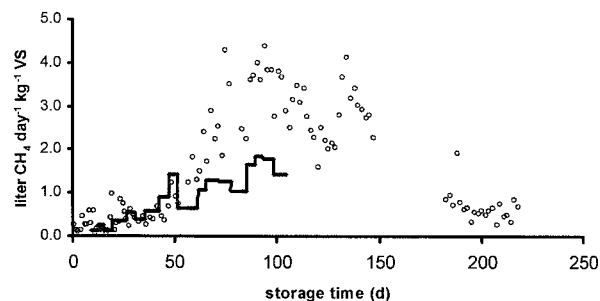
the other hand, if the CO₂ emissions measured are highest with the dynamic chamber technique, then this indicates either an aerobic degradation of organic components or oxidation of CH₄ in the surface layers. Figures 4 and 5 show that during storage of pig and cattle manure, the predominant emission of carbon was in the form of CO₂. Furthermore, the emissions of CH₄ estimated with the two techniques were within the same range, indicating that CH₄ oxidation was negligible, probably because the surface crust in the vessels had a high water content and CH₄ oxidation may have been low due to a low O₂ content of the crust. The emissions estimated with the static chamber technique only accounted for a minor part of the CO₂ emissions estimated by the dynamic chamber technique, indicating that there was a significant aerobic degradation of organic components in the surface layers of slurry in contact with the free atmosphere. Thus, in the following discussion, the ratio of anaerobic subsurface processes was estimated as the ratio of carbon-C emissions measured with the static chamber technique to that of the dynamic chamber technique.

In Fig. 6, the anaerobic carbon emission ratios are shown. In both pig and cattle manure, the degradation processes were mainly anaerobic for the first 20 d when stored at 15°C, but after this initial period there was a shift toward aerobic processes. When pig and cattle manure were stored at 20°C, the prevailing process was aerobic the first 20 d, but after this initial period, the

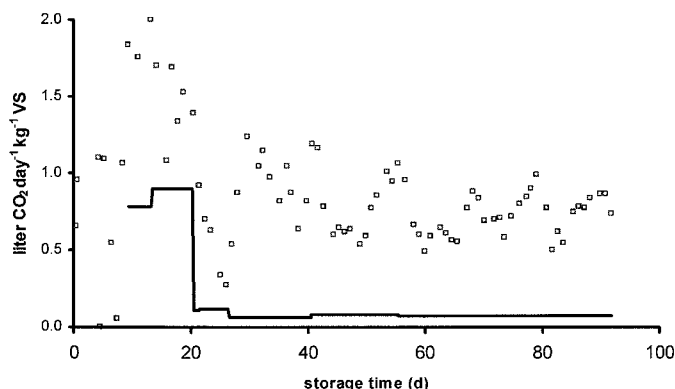
Pig manure stored at 15°C



Pig manure stored at 20°C



Pig manure stored at 15°C



Pig manure stored at 20°C

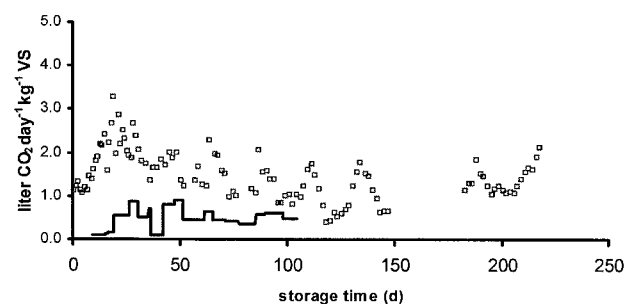


Fig. 4. Emissions of CO_2 and CH_4 from pig manure calculated with Eq. [1] and [2]. All registrations from subsurface and surface are average from 24 observations from three vessels. Emissions from only subsurface are average from three vessels during the period between emptying the static chamber. Circles, CH_4 emissions from surface and subsurface; squares, CO_2 emissions from surface and subsurface; solid line, CH_4 or CO_2 emissions from subsurface.

prevailing process shifted from being aerobic to being anaerobic. On average, the proportions of anaerobic emissions of carbon during the 90-d storage period were 31.2 and 38.2%, respectively, for pig and cattle manure stored at 15°C, whereas the proportion of anaerobic processes during the storage period was 39.0 and 58.0%, respectively, for pig and cattle manure stored at 20°C. Thus, the lower temperature (15°C) favored the aerobic processes, except for the initial period, while at the higher temperature (20°C), the aerobic and the anaerobic processes are of almost equal importance. In general, the anaerobic processes were more dominant during storage of cattle manure than during storage of pig manure.

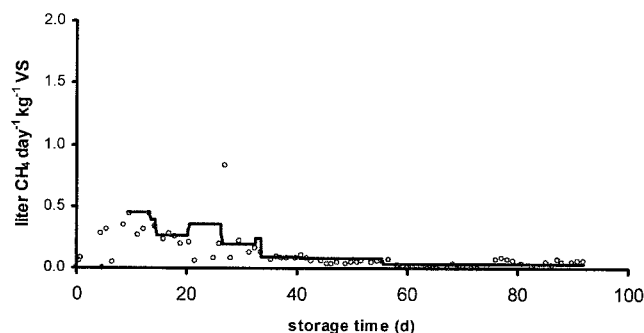
Accumulated Emissions

The cumulative loss of carbon during pre-storage of manure represents a loss in theoretical methane production (B_u). In practice, the percentage loss of the methane yield of the manure subsequent to storage [$L_{B_0}(t)$] was higher than the cumulative loss of carbon, since only

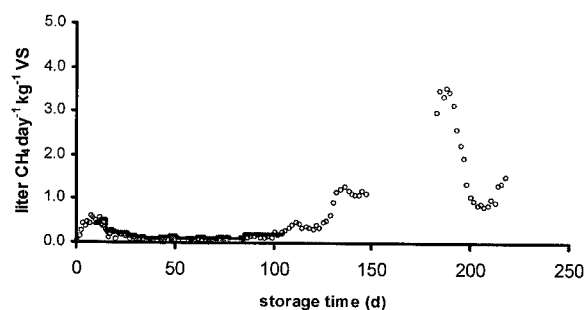
between 50 and 70% ($100B_0/B_u$) of the VS from cattle and pig manure was degradable during anaerobic degradation in a well-functioning digester (Table 1). After 5 d storage, the losses [$L_{B_0}(t)$] were between 1.8 and 3.8%, where the lowest loss was observed in pig manure stored at 15°C and the greatest loss was observed in cattle manure stored at 15°C (Fig. 7), illustrating that loss of methane production potential was not affected by temperature during the first days after storage of slurry. After 15 d storage, the losses [$L_{B_0}(t)$] were between 4.3 ± 0.6 and $6.6 \pm 0.4\%$, and the greatest loss was observed in cattle manure stored at 20°C. After 30 d, the loss had increased to between 7.7 ± 0.8 and $11.9 \pm 0.7\%$, and the greatest loss was still from cattle manure stored at 20°C. During long-term storage (90 d), the losses from pig manure stored at the higher temperature increased dramatically to about $41 \pm 6.7\%$ of the ultimate methane yield, whereas losses from cattle manure and pig manure stored at lower temperatures only increased moderately.

During the initial 30 d, the losses [$L_{B_0}(t)$] were $7.7 \pm$

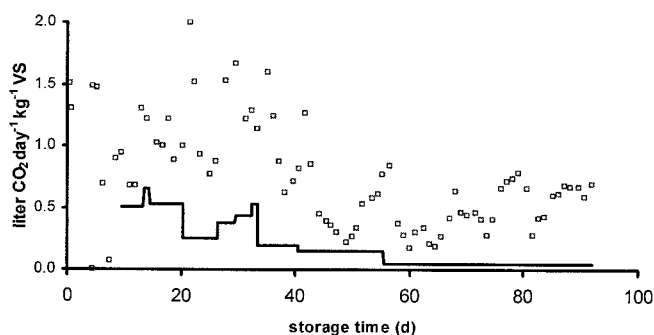
Cattle manure stored at 15°C



Cattle manure stored at 20°C



Cattle manure stored at 15°C



Cattle manure stored at 20°C

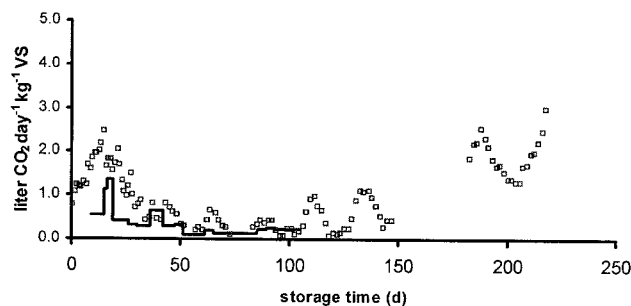


Fig. 5. Emissions of CO_2 and CH_4 from cattle manure calculated with Eq. [1] and [2]. All registrations from subsurface and surface are average from 24 observations from three vessels. Emissions from only subsurface are average from three vessels during the period between emptying the static chamber. Circles, CH_4 emissions from surface and subsurface; squares, CO_2 emissions from surface and subsurface; solid line, CH_4 or CO_2 emissions from subsurface.

0.8 and $10.9 \pm 2.1\%$ (Fig. 7 and 8) for pig manure stored at 15 and 20°C, respectively, while the corresponding MCF values were 3.7 ± 0.2 and $2.8 \pm 0.5\%$ (21.7 and $16.4 \text{ mg CH}_4 \text{ kg}^{-1} \text{ manure d}^{-1}$) when stored at 15 and 20°C in the same period (Table 2; Fig. 8). It thus seems that although the $L_{B_0}(t)$ increased at the high temperature, the MCF value was reduced as a consequence of a higher ratio of aerobic processes at the higher temperature in this initial phase. The losses $[L_{B_0}(t)]$ from cattle manure during the initial 30 d were 9.6 ± 0.8 and $11.9 \pm 0.7\%$ when stored at 15 and 20°C, while the corresponding MCF values were 2.6 ± 0.2 and $3.3 \pm 0.3\%$ (11.6 and $14.7 \text{ mg CH}_4 \text{ kg}^{-1} \text{ manure d}^{-1}$) when stored at 15 and 20°C in the same period. Thus, for cattle manure, slight increases in both MCF value and $L_{B_0}(t)$ were observed at the higher temperature during the initial 30 d.

After 90 d of storage, the losses $[L_{B_0}(t)]$ from pig manure kept at 15 and 20°C were 16.2 ± 2.2 and $41.8 \pm 6.7\%$, respectively, while the corresponding MCF values were 5.3 ± 0.5 and $31.3 \pm 5.4\%$ (9.3 and $55.1 \text{ mg CH}_4 \text{ kg}^{-1} \text{ manure d}^{-1}$) when stored at 15 and 20°C. Thus, both the accumulated carbon emission and the MCF value were strongly affected by the storage temperature during long-term storage of pig manure. After 90 d of

storage, the losses $[L_{B_0}(t)]$ from cattle manure kept at 15 and 20°C were 18.0 ± 2.8 and $17.0 \pm 2.6\%$, respectively, while the corresponding MCF values were 3.7 ± 0.3 and $4.3 \pm 0.9\%$ (5.5 and $6.4 \text{ mg CH}_4 \text{ kg}^{-1} \text{ manure d}^{-1}$) when stored at 15 and 20°C. Thus, both $L_{B_0}(t)$ and the MCF value were barely influenced by the temperature range of 15 to 20°C during a 90-d storage period for cattle manure. Külling et al. (2002) found CH_4 emissions in the range of 8.9 and $29.1 \text{ mg CH}_4 \text{ kg}^{-1} \text{ manure d}^{-1}$ during storage of cattle manure from cows fed different diets at 20°C for 100 d. Thus Külling et al. (2002) found higher emission rates than found in this study, which might be explained by the fact that that in their study CH_4 emissions were measured with a closed chamber technique without air exchange, which gives better conditions and more substrate for the methanogenic bacteria, since there is less substrate competition with aerobic bacteria.

Several studies have estimated CH_4 emissions during storage of liquid manure at temperatures between 10 and 30°C (Külling et al., 2002; Husted, 1994; Zeeman, 1994; Steed and Hashimoto, 1994), but all studies assume the degradation processes to be mainly anaerobic. In a closed manure system with no oxygen present, the

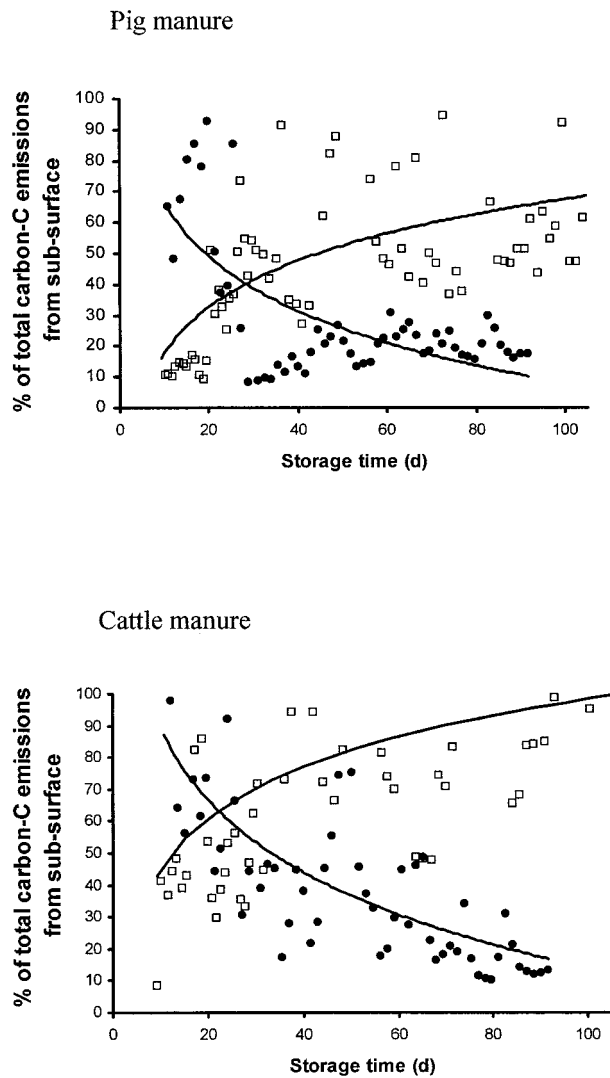


Fig. 6. Carbon emissions in the subsurface phase (mainly anaerobic) expressed in percentage of total emissions (anaerobic and aerobic emissions in the subsurface phase and in the surface layer). Anaerobic emissions (subsurface) are measured with Eq. [2], and total emissions are measured with Eq. [1]. Squares, 20°C; circles, 15°C.

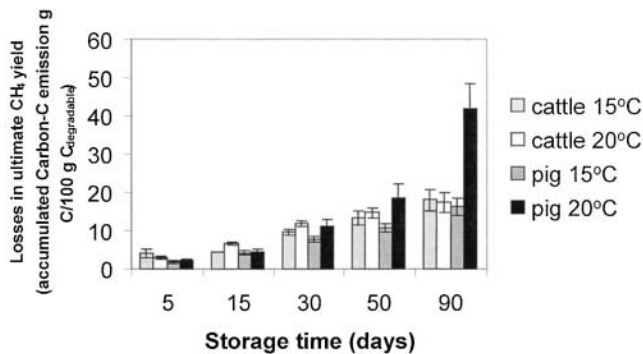


Fig. 7. The percentage losses in the ultimate methane yield that can be achieved from manure by anaerobic digestion $L_{B_0}(t)$. For the observations at Day 5, only 50% of the manure has been added, and this is taken into account in the calculations. Average retention time at Day 5 is 2.5 d. Error bars represent the standard deviation.

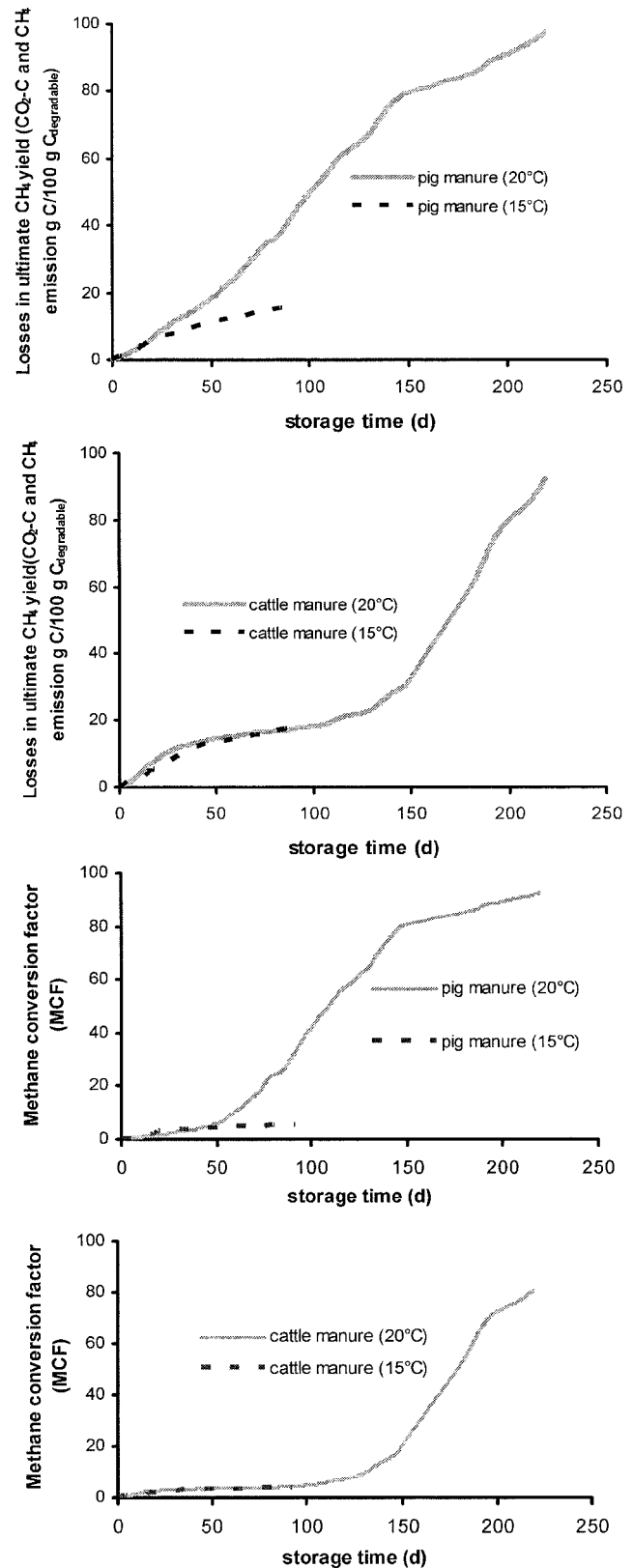


Fig. 8. The percentage losses in the ultimate methane yield that can be achieved from manure by anaerobic digestion $L_{B_0}(t)$ and methane conversion factor (methane recovered/methane potential B_0).

Table 2. Methane conversion factor (MCF) for the storage of pig and cattle manure at different temperatures and storage times.

Storage time	Reference	Storage temperature			
		15°C		20°C	
		Cattle manure	Pig manure	Cattle manure	Pig manure
d					
30	IPCC (1996) [†]	–	5	–	5
	this study [‡]	2.6 ± 0.15	3.7 ± 0.24	3.3 ± 0.33	2.8 ± 0.51
	Zeeman (1994) [§]	0	0	2	0
90	IPCC (1996) [†]	–	10	–	10
	this study [‡]	3.7 ± 0.28	5.3 ± 0.52	4.3 ± 0.94	31.3 ± 5.47
	Zeeman (1994) [§]	0	0	27	32
180	this study [‡]	–	–	53.0 ± 12.2	85.5 ± 12.8
	Zeeman (1994) [§]	27	41	41	45

[†] When manure is pre-stored inside buildings (pit storage), the Intergovernmental Panel on Climate Change (IPCC) only includes pig manure and distinguishes only between storage periods greater and less than 30 d. Methane emission (B_0) = 0.45 m³ kg⁻¹ volatile solids (VS) for pig manure and 0.24 m³ kg⁻¹ VS for cattle manure.

[‡] B_0 = 0.35 m³ kg⁻¹ VS for pig manure and 0.25 m³ kg⁻¹ VS for cattle manure.

[§] B_0 = 0.32 m³ kg⁻¹ VS for pig manure and 0.22 m³ kg⁻¹ VS for cattle manure.

observed MCF value is closely related to the loss in the potential methane production of the manure, and thus the MCF value expresses the reduction in the CH₄ yield that might be recovered by anaerobic digestion [$L_{B_0}(t)$] of the manure subsequent to pre-storage.

It is known that methanogenesis in manure strongly depends on the temperature when stored in a complete anaerobic system (Husted, 1994; Zeeman, 1994; Hill et al., 2001), but little is known about the influence on methanogenesis if the system is open to the atmosphere. Steed and Hashimoto (1994) found higher MCF values in closed systems than in systems that are open to the atmosphere, and explained this by inhibition of methanogenesis by oxygen. The results from the present study clearly illustrate a considerable methane and carbon dioxide production from pig and cattle manure in an open fed-batch system kept at 15 to 20°C, even at very low storage times, but the influence of temperature within this range on the MCF value were not very significant at storage times of <30 d. During long-term storage (90 d), there was a strong influence of temperature on the MCF value, especially for pig manure, where the MCF value increased from 5.3 ± 0.5% at the low temperature to 31.3 ± 5.5% at the high temperature, while correspondingly, the MCF value for cattle manure only increased from 3.7 ± 0.3 to 4.3 ± 0.9% at the higher temperature. The MCF values at storage times of <30 d calculated in this study are lower than those found by the Intergovernmental Panel on Climate Change (see Table 2), which could be explained by the fact that the system in this study is open to the atmosphere, which is known to inhibit methanogenesis.

CONCLUSIONS

The total content of VFA in manure increased within the first week after storage. The level of total VFA was higher in pig manure than in cattle manure. During the initial 90 d, acetic acid comprised more than 60% of the VFA in both cattle and pig manure incubated at 20°C, whereas in columns kept at 15°C, less than 60% of the VFA consisted of acetic acid. The relative content of propionate increased during the first 100 d of storage at both 15 and 20°C. The increase in propionate in stored

pig manure was higher than in cattle manure. The slurry pH increased during the first few days after the slurry application, due to degradation of urea to ammonia, but then decreased. The pH increased again after 50 d when the manure was kept at 20°C, while pH decreased further after more than 100 d when the manure was kept at 15°C.

At 15°C, aerobic processes in the surface layers of the stored slurry are more dominant than anaerobic processes, while at a higher temperature (20°C), the aerobic and anaerobic processes are of almost equal importance. In general, the anaerobic processes were more dominant during storage of cattle manure than during storage of pig manure. Manure was stored in open ventilated systems and a considerable part of the degradation of organic components was found to be aerobic; thus a greater reduction in the potential CH₄ production by anaerobic digestion subsequent to pre-storage than anticipated by the MCF value has been shown in this study, since the MCF value only includes anaerobic processes. Nevertheless, the MCF value is still the key parameter for evaluating the greenhouse gas emissions of the manure system, since CO₂ emissions from aerobic processes are not considered to contribute to global warming. The reduction in the potential CH₄ yield that might be recovered by anaerobic digestion of the manure subsequent to pre-storage is higher than indicated by the MCF value, because a considerable amount of the degradation processes were of the aerobic surface type. The losses in potential CH₄ yield from manure by anaerobic digestion after 5 d of storage were between 1.8 and 3.8%, where the lowest loss was observed in pig manure stored at 15°C and the highest loss was observed in cattle manure stored at 15°C. After 15 d storage, the loss was between 4.3 and 6.6%, where the highest loss was observed in cattle manure stored at 20°C, and after 30 d, the loss was between 7.7 and 11.9%. During long-term storage (>50 d), the losses from pig manure stored at the higher temperature will increase dramatically, whereas losses from cattle manure and pig manure stored at lower temperatures will only increase moderately.

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